CONFAI 2026 · PLAKSHA UNIVERSITY · TRACK 06 — RESPONSIBLE & ETHICAL AI

Refuse to Hallucinate

A deterministic validity firewall for Indian official survey statistics

Vidhi (विधि, *vidhi* — the method, the rule, the law) is a validity firewall for Indian official survey statistics. Every question gets a **cited verdict** — frozen with SHA-256 **before** any document is read. **StatsDoc** is MoSPI's document-search assistant (β): it finds numbers; Vidhi decides whether you may say them.

MoSPI's tools help you **find** the number. This firewall checks whether you are **allowed to say** it — and stops you, with a cited rule, when you are not.

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Poster + live demo · Live demo · localhost:8001/demo · offline · no API key · no GPU · $0.00 / query · Service · MCP tool · Agent · Library

1 The problem, and the fix

Retrieval over official PDFs is solved. **Governance is not.** Unconstrained RAG over sovereign statistics fails three ways — **two observed live** in MoSPI's StatsDoc pilot (*archived transcripts, chats 2472–2480*), the third general to RAG:

- **Inconsistent figures in one answer.** Measured MP MPCE quoted as both 3,158 and 5,589 for urban, and 5,538 vs 4,987 for "All", in a single response (chat 2474).
- **Unguided cross-round arithmetic.** ~40% MPCE growth quoted across the HCES 2011-12 → 2023-24 break (paper → 3-visit CAPI) with no caveat.
- **Retrieval poisoning.** A poisoned document chunk can override guardrails and flip a verdict. *(Not observed in the pilot — general to RAG; we test it against ourselves.)*

Why it matters. A journalist who prints 3,158 instead of 3,522 changes a state's consumption figure; a ministry repeats it. **The thesis.** You cannot fix a probabilistic system with more probabilistic prompts. The decision — *is this claim allowed?* — must come from **versioned, deterministic rules**, before generation.

The fix, end to end — a real response from the running service

"mpce for madhya pradesh"

VERDICT PASS_WITH_NUMBER RULE R_OK

3,522 Rs/month

Madhya Pradesh · rural · with imputation — and the slice is not narrowed: the open question returns **every published cell**, each with its own row id.

MP 2023-24	RURAL	URBAN	ALL-INDIA
with imputation	3,522	5,589	4,247
without imputation	3,441	5,538	4,122
StatsDoc said	3,158	5,011 / 5,589	4,987

```
source HCES 2023-24 factsheet, Statement 7
row HCES_2023_24_MPCE_2023-24_rural_with_MadhyaPradesh
verdict hash 7061f3ac… · tier VERIFIED
```

No number is shown without a row id, a PDF page and a hash.

EVERY REFUSAL IS CITED

A verdict is never bare prose — it carries a stable reason code.

- `R_OK` verified row matched
- `R_NUMERIC_MISMATCH` asserted number ≠ the store
- `R_SCOPE_OUT / R_FORECAST` forecast, causal, political → refused
- `R_NO_CURATED_ROW` not curated → coverage note, no figure
- `R_COMPARABILITY_*` survey break → mandatory caveat
- `R_PROMPT_INJECTION` injection refused before any lookup

2 The firewall

1. **Question** — natural language, any register
2. **Typed claim** — the translator: models translate, never judge
3. **The firewall** — five stages, fixed order — **the first rule to fire ends the answer**

S1	**Safety-A**	injection · EN / Hindi / Hinglish
S2	**Scope**	forecast · causal · politics
S3	**Comparability**	survey breaks
S4	**Metadata**	taxonomy check
S5	**Numeric**	verified rows only

↳ **the first rule that fires ends the answer** — a cited rule, and **no number**. All five must pass to continue.

4. **SHA-256 freeze — computed BEFORE retrieval**
   `hash = SHA-256(claim + decision + reason + rules)` — the hinge: after this line the verdict cannot change
5. **Retrieve** — untrusted document text — it cannot move the verdict
6. **Templated render + citations** — WARN/REFUSE wording is fixed

Safety-B · freeze before retrieval + figure attestation

The verdict hash is locked before any document is read. A planted "ignore the caveat" chunk changes nothing — the hash is byte-identical with or without poisoning. A second hash binds the **number you see** to its cell.

```
frozen_hash = SHA-256(claim ⊕ decision ⊕ sub_code ⊕ reason ⊕
rule_version)
attestation = SHA-256(frozen_hash ⊕ row_id ⊕ value ⊕ unit ⊕ pdf_sha256)
```

Field-guarded cache · no safe threshold

Danger-pair and paraphrase cosine ranges overlap on both measured embedders, so **no similarity threshold is safe**. Reuse only on exact critical-field match.

```
0 collisions / 19 danger pairs · 1708 verified rows
```

Generalised, not hard-coded

The translator resolves **every** state/UT the store carries, in any spelling — "mp", "up", "m.p.", "orissa", "maharastra" — one state or several at once. An open slice returns **every published cell**, never one narrowed figure.

```
"MPCE in Madhya Pradesh, UP, Maharashtra" → 12 cells · 12 row ids
```

What stops the three attacks

Prompt injection (EN · Hindi · Hinglish) — S1 refuses before any lookup, with a cited rule. **Poisoned chunk** — the hash is already frozen, so the chunk changes prose, never the verdict. **Cache poisoning** — exact critical-field match only, so a national row can never answer a state slice.

3 Results

0.000 hallucination rate · 36 must-abstain goldens	**1.000** abstention recall · 36 of 36 goldens
100 /100 verdict accuracy [0.963–1.000] · 60/60 numeric	**1.000** answer rate · 64 of 64 answerable, 0 refused
1400 /1400 rows re-verified on their PDF page	**1.000** Safety-A P/R/F1 · n=28, en · hi · hi-en
0.952 typed-claim accuracy · n=21, typos · codes · Hindi	**0.981** /0.785 cosine overlap · n=19 dangers, 6 paraphrases

VIDHI VS STATSDOC — HONEST COMPARISON

CHECK	STATSDOC (ARCHIVED)	VIDHI (MEASURED)
Consistency	3,158 vs 5,589 urban	one row id per figure
Cross-round math	~40% growth, no caveat	halts, cites the rule
Cost / latency	~$0.02–0.05 · 3.5–8 s *(est.)*	$0.00 · < 200 ms

StatsDoc rows: archived transcripts (chats 2472–2480, nine prompts). Vidhi rows: measured offline. One prompt set, one session — an observation, not a benchmark. Full table, including the rows not tested live, in `evaluation/statsdoc_audit/`.

THE STORE AT A GLANCE

Sources — PLFS Highlights 2023-24 & 2025, TUS 2024 factsheet & report, HCES 2023-24 factsheet & final report. Four official PDFs, committed and sha256-bound.

Rows — 1,708 verified; 1,397 (82%) also carry the PDF page they were read from, and 1,400 of 1,400 re-verify on that page.

Rules — version 4; six comparability rules with verbatim MoSPI quotes, and a freshness rule (store as of 2026-10-20). **Authored by the developer and not yet reviewed by a statistician** — that review is the first thing I would want next.

WHEN THE SYSTEM REFUSES

ASKED	WHAT HAPPENS
"MPCE in biharr"	abstains — "Did you mean **Bihar**?"
"MPCE in अफगानिस्तान"	abstains — never the all-India figure
"Is rural MPCE Rs 5,000?"	**R_NUMERIC_MISMATCH** — 3,522 is the row
"मध्य प्रदेश में MPCE?"	answers — Hindi resolves to the same row

HONEST LIMITS

- **Store as of 2026-10-20** — a later round is not in the store and says so, rather than answering "latest" with a stale row. **Curation is the real cost**: onboarding a survey is a manual, PDF-anchored job.
- **Coverage is the limit, not logic** — the PLFS Highlights battery, the TUS 2024 battery and the HCES 2023-24 state + trend columns. State-wise MPCE covers **all 36 states/UTs** in any spelling. Uncovered slices return `WARN_NO_DATA` + a coverage note, never a guess.
- Heuristic extraction breaks on exotic phrasing (an LLM translator port exists, `VIDHI_USE_LLM=1`, unmeasured); no load test yet.

First rule fires — two real traces from the running service

"Compare average MPCE in 2011-12 vs 2023-24."

S1 Safety-A	pass — no injection signal
S2 Scope	pass — descriptive
S3 Comparability	**WARN** R_COMPARABILITY_HCES_METHOD — 68th-round paper vs 3-visit CAPI

"Will unemployment fall in 2027?"

S1 Safety-A	pass — no injection signal
S2 Scope	**REFUSE** R_FORECAST — no such round exists

What the hash proves. In scope: reproducible and replayable — same question, same hash, and a planted chunk cannot move it. Out of scope: tamper-evidence against someone who holds the store. The attestation binds a *number* to a *row* — not that the row's reading of the PDF is the only sensible one.

Next. 1) A plain RAG+LLM baseline on the same 100 goldens and 28 red-team cases — the thesis is asserted today, not demonstrated. 2) A held-out set written by someone else. 3) Onboard CPI or ASI.

Six guarantees

1. Rules judge, models translate
2. Freeze before retrieval
3. No cosine threshold is safe
4. No PASS without provenance
5. Ambiguity → every cell, never one guess
6. Honest coverage — never a guess

REPRODUCE EVERYTHING (OFFLINE)

```
# no API key · no GPU · no network
pip install -r requirements.txt
python -m pytest tests/ -q        # 123 passed · every row reachable
python vidhi/eval.py               # 100 golden + danger + red-team + Safety-B
python -m uvicorn vidhi.service:app --port 8001  # → /demo
```

TRY THE LIVE DEMO

Live demo — `localhost:8001/demo`: ask a PLFS / HCES / TUS / MIS question. Watch the verdict freeze and every figure name its source row and page.

MCP: `vidhi.mcp_server` (4 governance tools) · **Agent:** `POST /analyze`.

Repo: `github.com/ajayc-iima/msai` · MIT · `v0.4` — 123 tests, the source PDFs and the StatsDoc transcripts are committed too.

THE QUESTION THIS WORK WON'T HALLUCINATE

Ask the hard question — expect `WARN_NO_DATA` + a coverage note, never a guess. Inject in English, Hindi or Hinglish and watch Safety-A refuse with a cited rule.

Questions and red-team attempts welcome.